

Reviewer Pack — Semialgebraic Dimension Bounds for SOS Approximation of Max-...

Entry 7443ec5addcc · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Semialgebraic Dimension Bounds for SOS Approximation of Max-CUT

Entry ID: 7443ec5addcc **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-19 10:27:07 UTC

1. Conjecture

Field A (mathematical branch): Real Algebraic Geometry **Field B** (complexity object): Sum-of-Squares Hierarchy

Statement:

For a random Max-CUT instance on n vertices, the minimal degree d required to achieve an α -approximation satisfies $d \geq \lceil \log(\chi(G)) \rceil$, where $\chi(G)$ is the chromatic number of the graph. This bound is preserved under polynomial-time reductions between Max-CUT instances.

Rationale (proposer’s reasoning):

Semialgebraic sets defined by polynomial inequalities constrain the SOS hierarchy’s ability to approximate Max-CUT. The chromatic number, a geometric invariant of the graph’s clique structure, dictates the minimal degree needed to capture the problem’s combinatorial complexity through real algebraic geometry.

Taxonomy category: SOS_DEGREE (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 488aec0533f65aef

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Generate a random Max-CUT instance on n=40 vertices
    n = 40
    G = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]
    for i in range(n):
        G[i][i] = 0

    # Compute the chromatic number  $\chi(G)$ 
    def color_graph(G):
        n = len(G)
        colors = [-1] * n
        available_colors = [set(range(2, n + 2)) for _ in range(n)]

        for i in range(n):
            if colors[i] == -1:
                used_colors = set()
                for j in range(i):
                    if G[i][j]:
                        used_colors.add(colors[j])
                available_colors[i] -= used_colors
                colors[i] = min(available_colors[i])

        return max(colors) + 1

     $\chi_G$  = color_graph(G)

    # Compute the minimal degree d required to achieve an  $\alpha$ -approximation
     $\alpha$  = 0.878
    d = math.ceil(math.log( $\chi_G$ ))

    if d == 0:
        return {
            "metric_name": "SOS_approximation_ratio",
```

```

        "metric_value": None,
        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

# Compute the SOS hierarchy's approximation ratio for degree d
def sos_hierarchy(G, d):
    n = len(G)
    A = [[0] * n for _ in range(n)]
    for i in range(n):
        for j in range(i + 1, n):
            if G[i][j]:
                A[i][j] = A[j][i] = 1

# Gaussian elimination to find the rank of A
def gaussian_elimination(A):
    m, n = len(A), len(A[0])
    for i in range(m):
        max_row = i
        for j in range(i + 1, m):
            if abs(A[j][i]) > abs(A[max_row][i]):
                max_row = j
        A[i], A[max_row] = A[max_row], A[i]

        if A[i][i] == 0:
            continue

        for j in range(n):
            A[i][j] /= A[i][i]

        for k in range(m):
            if k != i and A[k][i] != 0:
                factor = -A[k][i]
                for j in range(n):
                    A[k][j] += factor * A[i][j]

    rank = sum(1 for row in A if any(row))
    return rank

rank_A = gaussian_elimination(A)

# Compute the approximation ratio
return rank_A / n

approximation_ratio = sos_hierarchy(G, d)

return {
    "metric_name": "SOS_approximation_ratio",
    "metric_value": approximation_ratio,
    "instances_tested": 1,
    "conjecture_holds": approximation_ratio >=  $\alpha$ ,
    "counterexample": "" if approximation_ratio >=  $\alpha$  else f"Approximation ratio {approximation_ratio} <  $\alpha$ "
}

if __name__ == "__main__":

```

```

import sys
seeds = [int(s) for s in sys.argv[1:]] or [2**i + 1 for i in range(5, 30)]

results = []
for seed in seeds:
    trial_result = run_trial(seed)
    print(f"TRIAL: {trial_result}")
    results.append(trial_result)

mean_value = sum(result["metric_value"] for result in results if result["metric_value"] is not None)
std_value = math.sqrt(sum((result["metric_value"] - mean_value)**2 for result in results if result["metric_value"] is not None) / len(results))
support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

if all(result["conjecture_holds"] for result in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
else:
    for result in results:
        if not result["conjecture_holds"]:
            counterexample = result["counterexample"]
            first_failing_seed = seed
            break

    print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

tio', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
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TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
RESULT: SUPPORTED mean=1.0 std=0.0 support_fraction=1.0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge | original judge: All 100% tested instances (n=1) satisfy the conjecture with perfect metric value 1.0, meeting the $\geq 80\%$ support threshold. | next: Test conjecture on larger graphs with varying chromatic numbers to validate generalization

11. Audit log (LLM calls)

Total LLM calls: 15

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	113757
2	propose	ollama_remote	qwen3:8b	0	0	58153
3	propose	ollama_local	qwen3:8b	0	0	146947
4	propose	ollama_remote	qwen3:8b	0	0	45115
5	propose	ollama_remote	qwen3:8b	0	0	115794
6	propose	ollama_remote	qwen3:8b	0	0	109454
7	preregistration	ollama_remote	qwen3:8b	0	0	27466
8	novelty	ollama_remote	qwen3:8b	0	0	24189
9	novelty	ollama_remote	qwen3:8b	0	0	18536
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12931
11	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9628
12	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7220
13	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12088
14	critic	ollama_remote	qwen3:8b	0	0	33344
15	judge	ollama_remote	qwen3:8b	0	0	18797

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 753420 ms total latency. Provider mix: {'ollama_remote': 14, 'ollama_local': 1}

(full prompt+response transcripts available in `research/audit/7443ec5addcc.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/7443ec5addcc.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/7443ec5addcc.tar.gz` (if generated)